

Integration Testing

- Presumes previously tested units
- Not system testing
- Tests functionality "between" unit and system levels
- Basis for test case identification?
- Emphasis shifts from "how to test" to "what to test"

Approaches to Integration Testing (“source” of test cases)

- **Functional Decomposition** (most commonly described in the literature)
 - Top-down
 - Bottom-up
 - Sandwich
 - “Big bang”
- **Call graph**
 - Pairwise integration
 - Neighborhood integration
- **Paths**
 - MM-Paths
 - Atomic System Functions

Basis of Integration Testing Strategies

- **Functional Decomposition**
applies best to procedural code
- **Call Graph**
applies to both procedural and object-oriented code
- **MM-Paths**
applies to both procedural and object-oriented code

Top-Down Integration: Testing flows from the highest-level modules downward to the lowest. It uses **Stubs** to simulate missing lower-level components.

Bottom-Up Integration: Testing starts with the lowest-level modules and moves upward. It uses **Drivers** to simulate missing higher-level control modules.

Sandwich Integration: A hybrid approach that tests from both the top and bottom layers simultaneously, eventually meeting at a designated middle layer.